LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 1: Clones: One Sprite, Many Copies

Module 1 | 45 minutes | Clones | Grades 3–5 · ages 9–11

Objective: Use clones to make many copies of a sprite that each run their own script, and delete them when they are done.

Success criteria

- I can make clones from a hidden original.
- I can give each clone its own behaviour.
- I can delete clones when they are done.

Key vocabulary

- **clone**: A copy of a sprite made while the program runs.
- **original**: The sprite that makes the clones.
- **factory**: A script that keeps making clones.
- **delete this clone**: Removes a clone when it has finished.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Photocopier game (5 min): one learner is the original holding instructions; each “clone” (classmate) gets a copy of the same instruction card and follows it on their own.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Make it rain stars: learners follow the steps.
14 min	**Perform**	Code Lab: star shower, Count the copies, Life of a clone
9 min	**Probe**	Code Lab: endless snowfall, activity
4 min	**Wrap-up**	Exit ticket: Which hat block starts a clone’s own script?

Unplugged warm-up

Photocopier game (5 min): one learner is the original holding instructions; each “clone” (classmate) gets a copy of the same instruction card and follows it on their own.

Teaching the idea

Clones are runtime copies of a sprite. The original typically hides and acts as a factory (create clone of myself); each clone runs “when I start as a clone” independently and should delete itself when finished. The green flag removes all clones; Scratch caps clones at 300.

Analogy: A cookie cutter: one shape (the sprite) makes many cookies (clones), each baked on its own.

Live demo: Build the factory without deleting clones; show stars piling up. Add delete this clone. Then ask why the original should hide.

Questions to ask

Ask	Listen for
Which script runs for each clone?	“when I start as a clone”.
Why hide the original?	It is the factory, not part of the scene.
What happens if clones never delete?	They pile up and slow the game.

Differentiation

Support
Provide the clone script; learners build the factory loop.

Stretch
Give each clone a random size or speed using pick random inside the clone script.

Common misconceptions

Misconception	How to fix it
Clones are separate sprites in the sprite list	Clones exist only while running; they share the original’s code.
“when green flag clicked” runs for clones	Clones start with “when I start as a clone”.
Deleting the original deletes clones	Each clone deletes itself; the green flag clears all.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which hat block starts a clone’s own script?

Homework
Write down three things in a game you like that could be clones, and how each one would start and end.

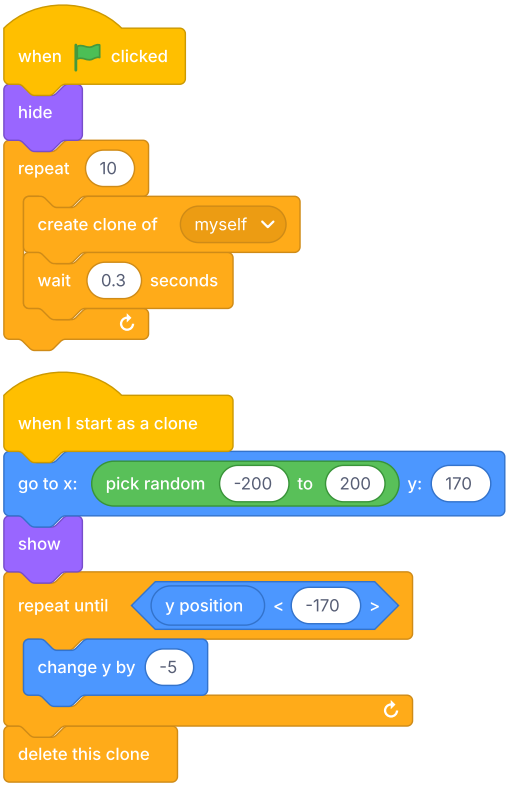
Answer key — Kitabash activities

1. Code Lab: star shower (perform · code lab)

Goals checked: Makes 10 clones · Clones start in random places · Clones clean up

One solution:

Star



2. Count the copies (perform · multiple choice)

Answer: 6

3. Life of a clone (perform · ordering)

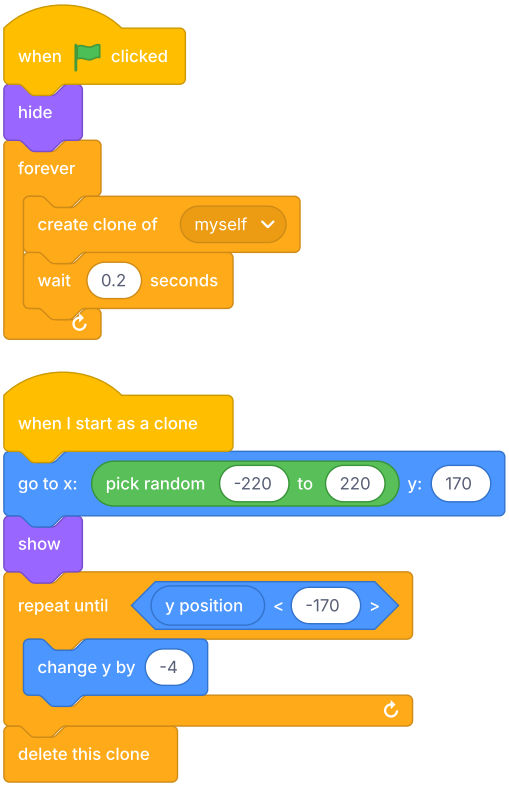
Order: The original runs “create clone of myself” → The clone starts its “when I start as a clone” script → The clone goes to a random place and shows itself → The clone falls until it reaches the bottom → The clone deletes itself

4. Code Lab: endless snowfall (probe · code lab)

Goals checked: Keeps making clones · Uses forever · Clones delete themselves

One solution:

Snow



5. Explain your thinking (probe · free response)

Look for: clone, many, copies. **Example:** Raindrops, enemies in a space game and coins in a platform game. Clones are better because one sprite can make as many copies as you need while the game runs, and you only write the code once.

Answer key — worksheet

1. **Key words:** clone = A copy of a sprite made while the program runs. · original = The sprite that makes the clones. · factory = A script that keeps making clones. · delete this clone = Removes a clone when it has finished.
2. **Plan your code:** Goals: Makes 10 clones; Clones start in random places; Clones clean up.
3. **Count the copies:** A. 6 — 1 original + 5 clones = 6. (Often the original is hidden.)
4. **Life of a clone:** 1 The original runs “create clone of myself” · 2 The clone starts its “when I start as a clone” script · 3 The clone goes to a random place and shows itself · 4 The clone falls until it reaches the bottom · 5 The clone deletes itself
5. **Plan your code:** Goals: Keeps making clones; Uses forever; Clones delete themselves.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.